

Antihypertensive Management in Acute Ischemic Stroke (AIS)

Blood Pressure Management in AIS

General Principles

- Elevated BP is **common and often compensatory** after acute stroke.
- Overly aggressive reduction in blood pressure can **worsen cerebral perfusion**—especially in the ischemic penumbra; avoid hypotension.
- BP targets differ **based on thrombolytic eligibility**.

BP Goals Based on Thrombolytic Use

Clinical Scenario	Treatment Threshold	Target BP
Eligible for thrombolytics (tPA/TNK)	Treat if SBP >185 mmHg or DBP >110 mmHg	Maintain <180/105 mmHg for 24h
Pre-thrombectomy and post-thrombolysis (first 24 hours)	Monitor and treat if SBP >180 or DBP >105	Maintain <180/105 mmHg
Not receiving thrombolytics or thrombectomy	Treat only if SBP >220 or DBP >120 mmHg	Reduce by ~15% over 24 hours
Hypertensive emergency (e.g., aortic dissection, MI, encephalopathy)	Immediate BP reduction regardless of thrombolytic use	Follow emergency guidelines

First-Line IV Antihypertensives

Medication	Dose	Notes
Labetalol	10–20 mg IV push; may repeat dose q10 min (max 80 mg per dose or 300 mg in 24 hours)	Rapid onset, good for bolus therapy if no pump available; avoid if suspected cocaine intoxication
Nicardipine	Start at 2.5–5 mg/hr IV infusion; titrate by 2.5 mg/hr every 5–15 min (max 15 mg/hr)	Preferred for continuous titration
Clevidipine	Start at 1–2 mg/hr, double every 2–5 min (max 21 mg/hr)	Rapid-on/offset; useful when tighter BP control needed
Hydralazine	10–20 mg IV q4–6h PRN	Variable response, longer onset

Avoid **nitroprusside or nitroglycerin** unless needed for other comorbidities—they may increase ICP.

References

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